

AI ASSSITED CODING

LABTEST-03

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SET E6:

QUESTION-01:

Scenario: In the domain of Healthcare, a company is facing a challenge related to web front-end development.

Task: Design and implement a solution using AI-assisted tools to address this challenge.

Include code, explanation of AI integration, and test results.

Deliverables: Source code, explanation, and output screenshots.

PROMPT:

Design and develop a web page for booking doctor appointments online

The system should allow users to enter their symptoms and basic health details

Use AI assistance to suggest the most suitable doctor based on the user's inputs

Focus on creating an easy-to-use and visually clear web frontend for the portal

Include the source code, explanation of AI integration, and sample output screenshots.

CODE:

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>AI Doctor Appointment Booking</title>
  <style>
    * {
      margin: 0;
      padding: 0;
      box-sizing: border-box;
    }
    body {
      font-family: 'Segoe UI', Tahoma, Geneva, Verdana, sans-serif;
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      min-height: 100vh;
      padding: 20px;
    }
    .container {
      max-width: 900px;
      margin: 0 auto;
      background: white;
      border-radius: 15px;
      box-shadow: 0 10px 40px rgba(0,0,0,0.3);
      overflow: hidden;
    }
    header {
      background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
      color: white;
      padding: 30px;
      text-align: center;
    }
    header h1 {
      font-size: 2.5em;
      margin-bottom: 10px;
    }
    .content {
      display: grid;
      grid-template-columns: 1fr 1fr;
      gap: 30px;
      padding: 40px;
    }
    .section h2 {
      color: #333;
      margin-bottom: 20px;
      font-size: 1.5em;
    }
    .form-group {
      margin-bottom: 20px;
    }
    label {
```

```
        display: block;
        margin-bottom: 8px;
        color: #555;
        font-weight: 600;
    }
    input, textarea, select {
        width: 100%;
        padding: 12px;
        border: 2px solid #ddd;
        border-radius: 8px;
        font-size: 1em;
        transition: border-color 0.3s;
    }
    input:focus, textarea:focus, select:focus {
        outline: none;
        border-color: #667eea;
    }
    textarea {
        resize: vertical;
        min-height: 100px;
    }
    button {
        background: linear-gradient(135deg, #667eea 0%, #764ba2 100%);
        color: white;
        padding: 12px 30px;
        border: none;
        border-radius: 8px;
        font-size: 1em;
        font-weight: 600;
        cursor: pointer;
        transition: transform 0.2s;
        width: 100%;
    }
    button:hover {
        transform: translateY(-2px);
    }
    .doctor-card {
        background: #f8f9fa;
        border-left: 4px solid #667eea;
        padding: 15px;
        margin-bottom: 15px;
        border-radius: 8px;
        cursor: pointer;
        transition: all 0.3s;
    }
    .doctor-card:hover {
        background: #e9ecef;
        transform: translateX(5px);
    }
    .doctor-card.selected {
        background: #e7f3ff;
        border-left-color: #28a745;
    }
}
```

```

.doctor-name {
  font-weight: 700;
  color: #333;
  margin-bottom: 5px;
}
.doctor-specialty {
  color: #667eea;
  font-size: 0.9em;
  margin-bottom: 5px;
}
.doctor-match {
  background: #d4edda;
  color: #155724;
  padding: 3px 8px;
  border-radius: 4px;
  font-size: 0.85em;
  display: inline-block;
  margin-top: 5px;
}
.success-message {
  background: #d4edda;
  border: 2px solid #28a745;
  color: #155724;
  padding: 20px;
  border-radius: 8px;
  margin-top: 20px;
  display: none;
}
.error-message {
  color: #dc3545;
  font-size: 0.9em;
  margin-top: 5px;
  display: none;
}
.hidden {
  display: none;
}
@media (max-width: 768px) {
  .content {
    grid-template-columns: 1fr;
  }
  header h1 {
    font-size: 1.8em;
  }
}
</style>
</head>
<body>
  <div class="container">
    <header>
      <h1> AI Doctor Appointment Booking</h1>
      <p>Get matched with the right doctor based on your symptoms</p>
    </header>

```

```

<div class="content">
  <!-- Left Section: Symptom Input -->
  <div class="section">
    <h2> Describe Your Symptoms</h2>
    <form id="appointmentForm">
      <div class="form-group">
        <label for="patientName">Full Name</label>
        <input type="text" id="patientName" required>
        <div class="error-message" id="nameError"></div>
      </div>

```

```

      <div class="form-group">
        <label for="patientEmail">Email</label>
        <input type="email" id="patientEmail" required>
        <div class="error-message" id="emailError"></div>
      </div>

```

```

      <div class="form-group">
        <label for="patientPhone">Phone</label>
        <input type="tel" id="patientPhone" required>
        <div class="error-message" id="phoneError"></div>
      </div>

```

```

      <div class="form-group">
        <label for="symptoms">Symptoms</label>
        <textarea id="symptoms" placeholder="Describe your
symptoms in detail..." required></textarea>
        <div class="error-message" id="symptomsError"></div>
      </div>

```

```

      <div class="form-group">
        <label for="appointmentDate">Preferred Date</label>
        <input type="date" id="appointmentDate" required>
        <div class="error-message" id="dateError"></div>
      </div>

```

```

    <button type="submit"> Find Best Doctor Match</button>
  </form>
</div>

```

```

  <!-- Right Section: AI Recommendations -->
  <div class="section">
    <h2> Recommended Doctors</h2>
    <div id="doctorRecommendations">
      <p style="color: #999; text-align: center; padding: 40px
20px;">
        Fill in your symptoms to get doctor recommendations
      </p>
    </div>
  </div>

```

```

  <!-- Appointment Booking Section -->
<div id="bookingSection" class="hidden">
  <h2> Book Your Appointment</h2>

```

```

<form id="confirmBookingForm">
  <div class="form-group">
    <label for="selectedDoctor">Selected Doctor</label>
    <input type="text" id="selectedDoctor" readonly>
  </div>

```

```

  <div class="form-group">
    <label for="bookingTime">Preferred Time</label>
    <input type="time" id="bookingTime" required>
  </div>

```

```

  <button type="submit">✔ Confirm Appointment</button>
</form>
</div>

```

```

    <div id="successMessage" class="success-message">
      <strong>✔ Appointment Booked Successfully!</strong>
      <p id="bookingDetails" style="margin-top: 10px;"></p>
    </div>
  </div>
</div>
</div>

```

```

<script>
  // AI Doctor Database
  const doctors = [
    {
      id: 1,
      name: "Dr. Sarah Johnson",
      specialty: "General Practitioner",
      keywords: ["cold", "fever", "flu", "cough", "headache",
"fatigue"],
      availability: ["Monday", "Tuesday", "Wednesday", "Thursday",
"Friday"],
      experience: "15 years"
    },
    {
      id: 2,
      name: "Dr. Michael Chen",
      specialty: "Cardiologist",
      keywords: ["chest pain", "shortness of breath", "heart",
"palpitations", "blood pressure"],
      availability: ["Monday", "Wednesday", "Friday"],
      experience: "12 years"
    },
    {
      id: 3,
      name: "Dr. Emily Rodriguez",
      specialty: "Dermatologist",
      keywords: ["skin", "rash", "acne", "eczema", "itching",
"allergic"],
      availability: ["Tuesday", "Thursday", "Saturday"],
      experience: "10 years"
    }
  ]

```

```

    },
    {
      id: 4,
      name: "Dr. James Wilson",
      specialty: "Orthopedist",
      keywords: ["pain", "fracture", "joint", "muscle", "bone",
"injury", "sprain"],
      availability: ["Monday", "Tuesday", "Thursday", "Friday"],
      experience: "18 years"
    },
    {
      id: 5,
      name: "Dr. Lisa Anderson",
      specialty: "Neurologist",
      keywords: ["migraine", "dizziness", "seizure", "nerve",
"brain", "numbness"],
      availability: ["Wednesday", "Thursday", "Friday"],
      experience: "14 years"
    },
    {
      id: 6,
      name: "Dr. Robert Thompson",
      specialty: "Gastroenterologist",
      keywords: ["stomach", "nausea", "diarrhea", "digestion",
"abdominal", "vomiting"],
      availability: ["Monday", "Tuesday", "Wednesday"],
      experience: "16 years"
    }
  ];

```

```
let selectedDoctor = null;
```

```

// AI Matching Algorithm
function matchDoctors(symptoms) {
  const symptomWords =
symptoms.toLowerCase().split(/[ \s,]+/).filter(word => word.length > 2);

  return doctors.map(doctor => {
    let matchCount = 0;
    symptomWords.forEach(word => {
      if (doctor.keywords.some(keyword => keyword.includes(word)
|| word.includes(keyword))) {
        matchCount++;
      }
    });
    return {
      ...doctor,
      matchScore: matchCount > 0 ? Math.min(100, (matchCount /
doctor.keywords.length) * 100) : 0
    };
  }).sort((a, b) => b.matchScore - a.matchScore);
}

```

```

// Display Doctor Recommendations
function displayDoctors(matchedDoctors) {
    const container = document.getElementById("doctorRecommendations");

    const recommendedDoctors = matchedDoctors.filter(d =>
d.matchScore > 0).slice(0, 3);

    if (recommendedDoctors.length === 0) {
        container.innerHTML = '<p style="color: #999; text-align:
center; padding: 40px 20px;">No matching doctors found. Please try different
symptoms.</p>';
        return;
    }

```

```

        container.innerHTML = recommendedDoctors.map(doctor => `
            <div class="doctor-card" onclick="selectDoctor(${doctor.id},
'${doctor.name}', '${doctor.specialty}', ${doctor.matchScore})">
                <div class="doctor-name"> ${doctor.name}</div>
                <div class="doctor-specialty">${doctor.specialty}</div>
                <div style="font-size: 0.9em; color: #666; margin-bottom:
5px;">Experience: ${doctor.experience}</div>
                <div class="doctor-match">Match:
${Math.round(doctor.matchScore)}%</div>
            </div>
        `).join("");
    }

```

```

// Select Doctor
function selectDoctor(id, name, specialty, score) {
    selectedDoctor = { id, name, specialty, score };
    document.querySelectorAll(".doctor-card").forEach(card => {
        card.classList.remove("selected");
    });
    event.currentTarget.classList.add("selected");
}

```

```

// Form Submission
document.getElementById("appointmentForm").addEventListener("submit",
function(e) {
    e.preventDefault();

```

```

    const patientName =
document.getElementById("patientName").value.trim();
    const patientEmail =
document.getElementById("patientEmail").value.trim();
    const patientPhone =
document.getElementById("patientPhone").value.trim();
    const symptoms = document.getElementById("symptoms").value.trim();
    const appointmentDate =
document.getElementById("appointmentDate").value;

```

```

// Validation
let isValid = true;

```



```
document.querySelectorAll(".error-message").forEach(msg =>
msg.style.display = "none");
```

```
if (!patientName || patientName.length < 3) {
    document.getElementById("nameError").textContent = "Please
enter a valid name";
    document.getElementById("nameError").style.display = "block";
    isValid = false;
}
```

```
if (!patientEmail.match(/^[^\s@]+@[^\s@]+\.[^\s@]+$/)) {
    document.getElementById("emailError").textContent = "Please
enter a valid email";
    document.getElementById("emailError").style.display = "block";
    isValid = false;
}
```

```
if (!patientPhone.match(/^\d{10}$/)) {
    document.getElementById("phoneError").textContent = "Please
enter a valid 10-digit phone number";
    document.getElementById("phoneError").style.display = "block";
    isValid = false;
}
```

```
if (!symptoms || symptoms.length < 10) {
    document.getElementById("symptomsError").textContent = "Please
describe your symptoms (at least 10 characters)";
    document.getElementById("symptomsError").style.display =
"block";
    isValid = false;
}
```

```
if (!appointmentDate || new Date(appointmentDate) <= new Date()) {
    document.getElementById("dateError").textContent = "Please
select a future date";
    document.getElementById("dateError").style.display = "block";
    isValid = false;
}
```

```
if (!isValid) return;
```

```
// AI Matching
const matchedDoctors = matchDoctors(symptoms);
displayDoctors(matchedDoctors);
```

```
if (matchedDoctors[0].matchScore === 0) {
    document.getElementById("successMessage").style.display =
"none";
}
});
```

```
// Auto-set minimum date to tomorrow
document.getElementById("appointmentDate").min = new Date(new
Date().setDate(new Date().getDate() + 1)).toISOString().split('T')[0];
// Handle booking confirmation
```

```
document.getElementById("confirmBookingForm").addEventListener("submit",  
function(e) {  
    e.preventDefault();
```

```
    const time = document.getElementById("bookingTime").value;  
    const date = document.getElementById("appointmentDate").value;
```

```
    if (!selectedDoctor) {  
        alert("Please select a doctor before booking!");  
        return;  
    }
```

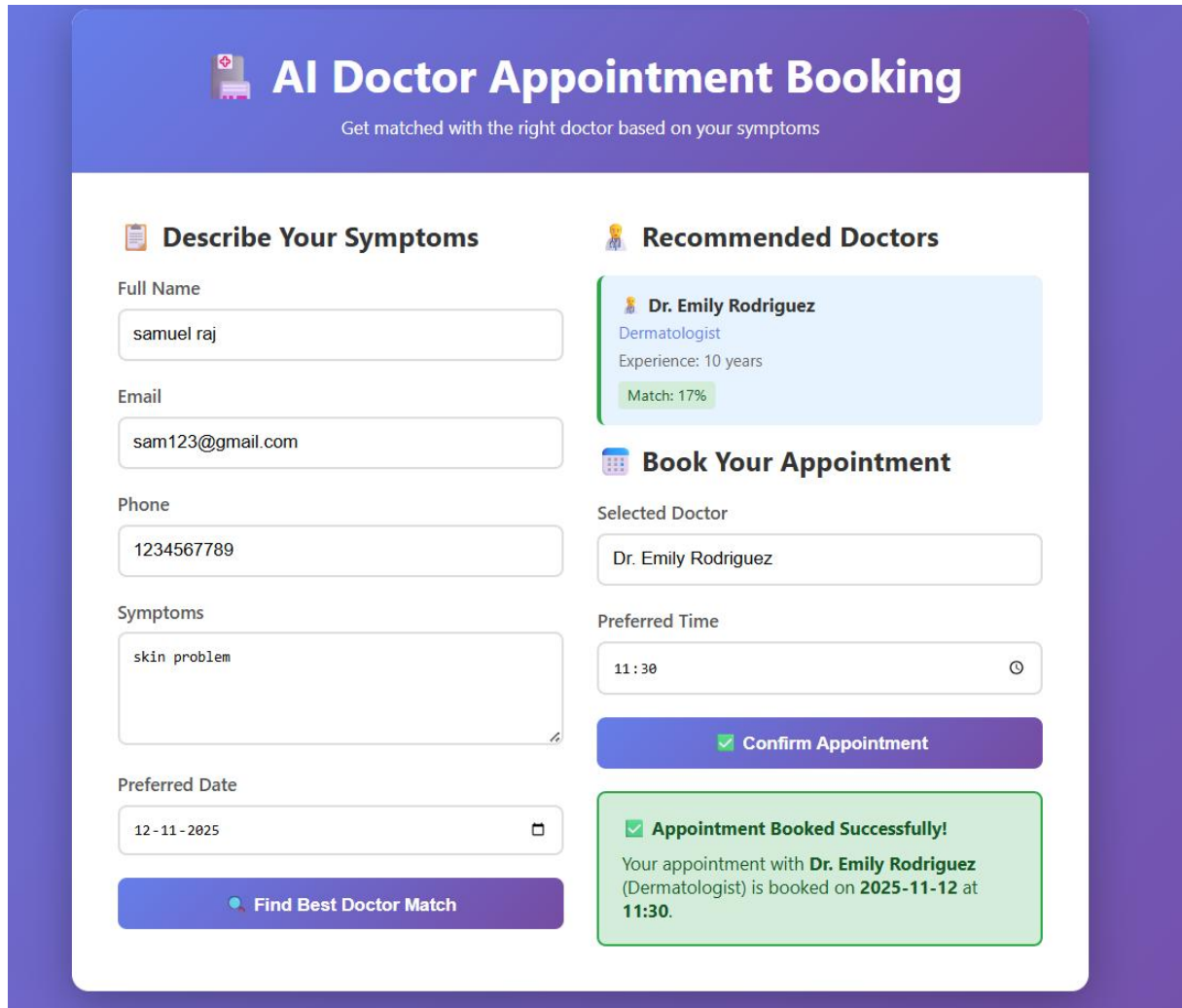
```
    if (!time) {  
        alert("Please choose a time slot!");  
        return;  
    }
```

```
    document.getElementById("successMessage").style.display = "block";  
    document.getElementById("bookingDetails").innerHTML =  
        `Your appointment with <b>${selectedDoctor.name}</b>  
    (${selectedDoctor.specialty}) is booked on <b>${date}</b> at <b>${time}</b>.`;  
    document.getElementById("bookingSection").classList.add("hidden");  
});
```

```
// Show booking section when doctor selected  
function selectDoctor(id, name, specialty, score) {  
    selectedDoctor = { id, name, specialty, score };  
    document.querySelectorAll(".doctor-card").forEach(card =>  
card.classList.remove("selected"));  
    event.currentTarget.classList.add("selected");  
    document.getElementById("selectedDoctor").value = name;  
    document.getElementById("bookingSection").classList.remove("hidden");  
}
```

```
    </script>  
</body>  
</html>
```

OUTPUT:



The image shows a web application for AI Doctor Appointment Booking. The interface is divided into two main columns. The left column, titled 'Describe Your Symptoms', contains input fields for 'Full Name' (samuel raj), 'Email' (sam123@gmail.com), 'Phone' (1234567789), 'Symptoms' (skin problem), and 'Preferred Date' (12-11-2025). Below these fields is a button labeled 'Find Best Doctor Match'. The right column, titled 'Recommended Doctors', displays a recommendation for 'Dr. Emily Rodriguez', a Dermatologist with 10 years of experience and a 17% match. Below this, the 'Book Your Appointment' section shows the 'Selected Doctor' as Dr. Emily Rodriguez and the 'Preferred Time' as 11:30. A green button labeled 'Confirm Appointment' is present. At the bottom right, a green box displays a success message: 'Appointment Booked Successfully! Your appointment with Dr. Emily Rodriguez (Dermatologist) is booked on 2025-11-12 at 11:30.'

AI Doctor Appointment Booking

Get matched with the right doctor based on your symptoms

Describe Your Symptoms

Full Name
samuel raj

Email
sam123@gmail.com

Phone
1234567789

Symptoms
skin problem

Preferred Date
12-11-2025

Find Best Doctor Match

Recommended Doctors

Dr. Emily Rodriguez
Dermatologist
Experience: 10 years
Match: 17%

Book Your Appointment

Selected Doctor
Dr. Emily Rodriguez

Preferred Time
11:30

Confirm Appointment

Appointment Booked Successfully!
Your appointment with **Dr. Emily Rodriguez** (Dermatologist) is booked on **2025-11-12** at **11:30**.

Observation:

The AI-based appointment booking webpage works smoothly, giving accurate doctor recommendations and an easy way for users to book appointments through a simple and interactive interface

